

Econometrics: Enfoques Modernos en Econometría y Ciencia de Datos Investigación y Aplicación



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Investigación y desarrollo presentada por

Econometrics: Econometría con Enfoques Modernos

Quito, Ecuador

A mi familia, mentores y a mis queridos estudiantes por su apoyo continuo e
inspiración.

Agradecimientos

Expreso mi más sincero agradecimiento a mi madre, Azucena Manosalvas, a mis hermanos y a mi querida amiga Jhennyfer Eivar, cuyo apoyo y compañía han sido fundamentales en mi camino. A Jhennyfer, en particular, le agradezco por ayudarme a mantener la mirada en la pragmática de la vida, recordándome que el conocimiento adquiere verdadero sentido cuando puede comprenderse, aplicarse y aportar a la realidad.

A mis mentores, gracias por compartir sus conocimientos, exigirme siempre un poco más y enseñarme que aprender también significa cuestionar, explorar y construir nuevas formas de comprender el mundo. A mis queridos estudiantes, mi especial gratitud: ustedes me han enseñado que toda explicación debe partir de la sencillez, de esa manera en que un niño observa su entorno con curiosidad, sin complicaciones innecesarias y sin abrumar su propio proceso de descubrimiento. Gracias por recordarme que, muchas veces, comprender lo complejo comienza por saber explicarlo de manera sencilla.

Extiendo también mi profundo agradecimiento a la Universidad de las Fuerzas Armadas ESPE, por su guía continua, su rigurosidad académica y las valiosas enseñanzas que han acompañado mi formación profesional y personal.

Finalmente, agradezco de corazón a mis seres queridos por su apoyo incondicional, su paciencia y su confianza en cada etapa del camino. Cada palabra de aliento, cada gesto de comprensión y cada momento compartido.

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Resumen

Esta investigación presenta la aplicación de enfoques modernos en econometría y ciencia de datos, integrando herramientas computacionales avanzadas en *R Markdown* y **bookdown** para el análisis cuantitativo riguroso, reproducible y aplicado a los negocios y pequeñas y medianas empresas (PYMES).

El trabajo aborda el modelado estadístico y econométrico mediante flujos de trabajo computacionales transparentes, permitiendo vincular directamente fuentes de datos, estimaciones econométricas, visualizaciones dinámicas y documentación teórica en un solo entorno reproducible.

A través del proyecto **Econometrics**, se promueve la adopción de metodologías empíricas sólidas que buscan contribuir a la transformación de la gestión organizacional y facilitar la toma de decisiones basada en evidencia cuantitativa. Los desarrollos, investigaciones y soluciones vinculadas al proyecto se encuentran disponibles en econometricis.vercel.app.

Palabras clave: Econometría, ciencia de datos, análisis cuantitativo, PYMES, R Markdown, *bookdown*, reproducibilidad, toma de decisiones basada en evidencia.

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Lista de Abreviaturas

1-D, 2-D	. . .	Una o dos dimensiones, correspondiente a las dimensiones espaciales o matriciales del análisis.
ESPE		Universidad de las Fuerzas Armadas ESPE.
PYMES		Pequeñas y medianas empresas.
MCO		Mínimos Cuadrados Ordinarios.

Introducción

Bienvenido a la investigación y marco computacional de **EconometricSI** — **Enfoques Modernos**, desarrollado por **Dilan Alexander Manosalvas Andrade** en la **Universidad de las Fuerzas Armadas ESPE**.

Este entorno permite redactar documentos académicos, tesis e investigaciones de alto impacto utilizando **R Markdown** y **bookdown**, integrando de manera fluida el análisis econométrico, modelos estadísticos y visualizaciones avanzadas con una presentación editorial rigurosa tanto en formato digital (HTML) como impreso (PDF) y procesadores de texto (Word).

Escribir investigaciones en R Markdown permite embeber bloques de código (*chunks*) directamente en el texto del informe, generando tablas y figuras dinámicamente a partir de los microdatos subyacentes sin recurrir a procesos manuales de copiar y pegar. Esto fomenta la investigación reproducible y científica, facilitando la auditoría, replicación y mejora de los resultados empíricos obtenidos.

¿Por qué utilizar este enfoque?

R Markdown ofrece una interfaz ágil, moderna e intuitiva para aprovechar la potencia tipográfica de LaTeX:

- Los paquetes de **R** y **Python** interactúan directamente para generar modelos econométricos, tablas estilizadas y gráficos de alta calidad.
- Permite leer datos, limpiarlos, estimar especificaciones econométricas y pasar resultados directamente al cuerpo del texto en tiempo real mediante código en línea.

Introducción

- Genera flujos de trabajo reproducibles donde cualquier actualización en los datos recalcula automáticamente todas las estadísticas, tablas y gráficos.
- Brinda estabilidad y consistencia editorial superior frente a procesadores de palabras convencionales, manteniendo archivos fuente ligeros y versionables con Git y GitHub.

¿A quién está dirigida esta investigación?

Esta obra está dirigida a investigadores, economistas, científicos de datos, estudiantes y profesionales del área empresarial y cuantitativa interesados en aplicar enfoques modernos de econometría, análisis de datos y modelación estadística a la toma de decisiones estratégicas, optimización operativa y crecimiento de pequeñas y medianas empresas (PYMES).

1

Guía de Uso del Entorno

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1.1. Estructura del Proyecto

```
.
+-- index.Rmd
+-- _bookdown.yml
+-- 00-introduction.Rmd
/   ...
+-- 07-conclusion.Rmd
+-- front-and-back-matter
/   +-- _abstract.Rmd
/   +-- 98-appendices.Rmd
/   ...
+-- bibliography
/   +-- references.bib
/   ...
```

1. Guía de Uso del Entorno

```
+-- figures
/   ...
+-- docs
/   +-- _main.pdf
/   ...
+-- scripts_and_filters
/   +-- knit-functions.R
/   ...
+-- templates
/   +-- template.tex
/   ...
```

1.1.1. index.Rmd: metadatos y opciones de diseño

En `index.Rmd` se definen los metadatos principales del documento (título, autor, institución):

```
title: |
  EconometricSI: \
  Enfoques Modernos en Econometría y Ciencia de Datos \
  Investigación y Aplicación
author: Dilan Alexander Manosalvas Andrade
college: Departamento de Ciencias Económicas, Administrativas y de
↪ Comercio
university: Universidad de las Fuerzas Armadas ESPE
```

También se configuran las rutas hacia el resumen, agradecimientos, abreviaturas y bibliografía (uno o varios archivos **.bib**):

```
abstract: |
  Esta investigación presenta la aplicación de enfoques modernos en
↪ econometría y ciencia de datos, integrando herramientas
↪ computacionales avanzadas en *R Markdown* y `bookdown` para el
↪ análisis cuantitativo riguroso, reproducible y aplicado a los
↪ negocios y pequeñas y medianas empresas (PYMES).
```

```
El trabajo aborda el modelado estadístico y econométrico mediante
↪ flujos de trabajo computacionales transparentes, permitiendo
↪ vincular directamente fuentes de datos, estimaciones
↪ econométricas, visualizaciones dinámicas y documentación teórica
↪ en un solo entorno reproducible.
```

```
A través del proyecto Econometrics, se promueve la adopción de
↪ metodologías empíricas sólidas que buscan contribuir a la
↪ transformación de la gestión organizacional y facilitar la toma
↪ de decisiones basada en evidencia cuantitativa. Los desarrollos,
↪ investigaciones y soluciones vinculadas al proyecto se
↪ encuentran disponibles en
```

1. Guía de Uso del Entorno

****Palabras clave:**** Econometría, ciencia de datos, análisis
→ cuantitativo, PYMES, R Markdown, *bookdown*, reproducibilidad,
→ toma de decisiones basada en evidencia.

acknowledgements: |

Expreso mi más sincero agradecimiento a mi madre, Azucena
→ Manosalvas, a mis hermanos y a mi querida amiga Jhenyfer Eivar,
→ cuyo apoyo y compañía han sido fundamentales en mi camino. A
→ Jhenyfer, en particular, le agradezco por ayudarme a mantener
→ la mirada en la pragmática de la vida, recordándome que el
→ conocimiento adquiere verdadero sentido cuando puede
→ comprenderse, aplicarse y aportar a la realidad.

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→ siempre un poco más y enseñarme que aprender también significa
→ cuestionar, explorar y construir nuevas formas de comprender el
→ mundo. A mis queridos estudiantes, mi especial gratitud: ustedes
→ me han enseñado que toda explicación debe partir de la
→ sencillez, de esa manera en que un niño observa su entorno con
→ curiosidad, sin complicaciones innecesarias y sin abrumar su
→ propio proceso de descubrimiento. Gracias por recordarme que,
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→ explicarlo de manera sencilla.

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→ formación profesional y personal.

Finalmente, agradezco de corazón a mis seres queridos por su apoyo
→ incondicional, su paciencia y su confianza en cada etapa del
→ camino. Cada palabra de aliento, cada gesto de comprensión y
→ cada momento compartido.

`\begin{flushright}`
Dilan Alexander Manosalvas Andrade \\
Universidad de las Fuerzas Armadas ESPE \\
Quito, Ecuador
`\end{flushright}`

dedication: A mi familia y mentores

abbreviations: |

`\item[1-D, 2-D]` Una o dos dimensiones, correspondiente a las
→ dimensiones espaciales o matriciales del análisis.

1. Guía de Uso del Entorno

```
\item[ESPE] Universidad de las Fuerzas Armadas ESPE.  
  
\item[PYMES] Pequeñas y medianas empresas.  
  
\item[MCO] Mínimos Cuadrados Ordinarios.  
  
#####  
## bibliography path ##  
#####  
bibliography: [bibliography/references.bib,  
↪ bibliography/additional-references.bib]
```

Finalmente, en **index.Rmd** se personalizan las opciones de salida: encabezados de bibliografía, numeración de páginas, idioma (Español) y estilos visuales.

1.1.2. Otros archivos .Rmd en la raíz: Capítulos

- Cada capítulo cuenta con su propio archivo **.Rmd** en el directorio raíz.
- Al compilar **index.Rmd**, los capítulos se combinan automáticamente en orden alfanumérico según sus nombres de archivo.

1.1.3. front-and-back-matter/

- Carpeta que contiene las secciones preliminares y finales.
- Los archivos con guión bajo inicial (como **__abstract.Rmd**) no se fusionan automáticamente, sino que son llamados explícitamente desde **index.Rmd**.
- Los archivos como **98-appendices.Rmd** y **99-references_heading.Rmd** se integran al final del documento.

1.1.4. templates

- **template.tex**: plantilla LaTeX para compilar el documento completo a PDF.
- **brief_template.tex**: plantilla para compilar previsualizaciones de capítulos individuales.
- **beltcrest.pdf**: logotipo institucional (Universidad de las Fuerzas Armadas ESPE) de la portada.

1.2. Compilación del Documento Completo

- Para compilar el documento completo, abra **index.Rmd** en RStudio y haga clic en el botón **Knit**.
- Los archivos compilados se guardan automáticamente en la carpeta **docs/** (ideal para GitHub Pages).
- Para elegir el formato de salida, en el encabezado de **index.Rmd** configure `thesis_formats <- "pdf"`; o combine múltiples formatos como `c("pdf", "bs4", "gitbook", "word")`.

Salida en PDF

```
knit: (function(input, ...) {  
  thesis_formats <- "pdf";  
  ...  
})
```

Salida BS4 book (HTML interactivo)

```
knit: (function(input, ...) {  
  thesis_formats <- "bs4";  
  ...  
})
```

Salida GitBook (HTML)

```
knit: (function(input, ...) {  
  thesis_formats <- "gitbook";  
  ...  
})
```

Salida en Word (.docx)

```
knit: (function(input, ...) {  
  thesis_formats <- "word";  
  ...  
})
```

Neque porro quisquam est qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit...

There is no one who loves pain itself, who seeks after it and wants to have it, simply because it is pain...

— Cicero's *de Finibus Bonorum et Malorum*.

2

R Markdown basics

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Here is a brief introduction to using *R Markdown*. *Markdown* is a simple formatting

2. *R Markdown basics*

syntax for authoring HTML, PDF, and MS Word documents and much, much more. *R Markdown* provides the flexibility of *Markdown* with the implementation of **R** input and output. For more details on using *R Markdown* see <http://rmarkdown.rstudio.com>.

2.1. Basic markdown syntax

2.1.1. Whitespace

Be careful with your spacing. While whitespace largely is ignored, it does at times give markdown signals as to how to proceed. As a habit, try to keep everything left aligned whenever possible, especially as you type a new paragraph. In other words, there is no need to indent basic text in the Rmd document (in fact, it might cause your text to do funny things if you do).

2.1.2. Italics and bold

- *Italics* are done like `*this*` or `_this_`
- **Bold** is done like `**this**` or `__this__`
- ***Bold and italics*** is done like `***this***`, `____this____`, or (the most transparent solution, in my opinion) `**_this_**`

2.1.3. Inline code

- Inline code is created with backticks like ``this``

2.1.4. Sub and superscript

Sub₂ and super² script is created like `this~2~` and `this^2^`

2.1.5. Strikethrough

- stStrikethrough is done `~~like this~~`

2.1.6. ‘Escaping’ (aka “What if I need an actual asterisk?”)

- To include an actual *, _ or \, add another \ in front of them: *, _, \\

2.1.7. Endash (–), emdash (—)

- – and — with -- and ---

2.1.8. Blockquotes

Do like this:

Put a > in front of the line.

2.1.9. Headings

Section headers are created with #'s of increasing number, i.e.

- # First-level heading
- ## Second-level heading
- ### Etc.

In PDF output, a level-five heading will turn into a paragraph heading, i.e. `\paragraph{My level-five heading}`, which appears as bold text on the same line as the subsequent paragraph.

2.1.10. Lists

Unordered list by starting a line with an * or a -:

- Item 1
- Item 2

Ordered lists by starting a line with a number. Notice that you can mislabel the numbers and *Markdown* will still make the order right in the output:

1. Item 1

2. *R Markdown basics*

2. Item 2

To create a sublist, indent the values a bit (at least four spaces or a tab):

1. Item 1
2. Item 2
3. Item 3
 - Item 3a
 - Item 3b

2.1.11. Line breaks

The official *Markdown* way to create line breaks is by ending a line with more than two spaces.

Roses are red. Violets are blue.

This appears on the same line in the output, because we didn't add spaces after red.

Roses are red.

Violets are blue.

This appears with a line break because I added spaces after red.

I find this is confusing, so I recommend the alternative way: Ending a line with a backslash will also create a linebreak:

Roses are red.

Violets are blue.

To create a new paragraph, you put a blank line.

Therefore, this line starts its own paragraph.

2.1.12. Hyperlinks

- [This is a hyperlink](https://hyperlink-in-parentheses) created by writing the text you want turned into a clickable link in [square brackets followed by a](https://hyperlink-in-parentheses)

2.1.13. Footnotes

- Are created¹ by writing either `^[my footnote text]` for supplying the footnote content inline, or something like `[^a-random-footnote-label]` and supplying the text elsewhere in the format shown below²:

`[^a-random-footnote-label]: This is a random test.`

2.1.14. Comments

To write comments within your text that won't actually be included in the output, use the same syntax as for writing comments in HTML. That is, `<!-- this will not be included in the output -->`.

2.1.15. Math

The syntax for writing math is stolen from LaTeX. To write a math expression that will be shown **inline**, enclose it in dollar signs. - This: `$A = \pi*r^{2}$`
Becomes: $A = \pi * r^2$

To write a math expression that will be shown in a block, enclose it in two dollar signs.

This: `$$A = \pi*r^{2}$$`

Becomes:

$$A = \pi * r^2$$

To create numbered equations, put them in an 'equation' environment and give them a label with the syntax `(\#eq:label)`, like this:

```
\begin{equation}
  f\left(k\right) = \binom{n}{k} p^k\left(1-p\right)^{n-k}
  (\#eq:binom)
\end{equation}
```

¹my footnote text

²This is a random test.

2. R Markdown basics

Becomes:

$$f(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad (2.1)$$

For more (e.g. how to theorems), see e.g. the documentation on bookdown.org

2.2. Executable code chunks

The magic of R Markdown is that we can add executable code within our document to make it dynamic.

We do this either as *code chunks* (generally used for loading libraries and data, performing calculations, and adding images, plots, and tables), or *inline code* (generally used for dynamically reporting results within our text).

The syntax of a code chunk is shown in Figure 2.1.

```
## Warning: package 'tidyverse' was built under R version 4.4.3
## Warning: package 'ggplot2' was built under R version 4.4.3
## Warning: package 'tidyr' was built under R version 4.4.3
## Warning: package 'purrr' was built under R version 4.4.3
## Warning: package 'dplyr' was built under R version 4.4.3
## Warning: package 'lubridate' was built under R version 4.4.3
```

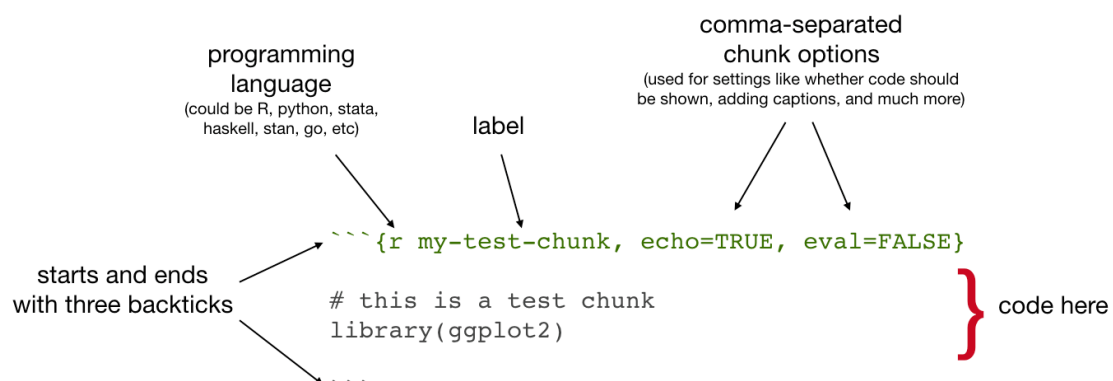


Figure 2.1: Code chunk syntax

Common chunk options include (see e.g. bookdown.org):

2. R Markdown basics

- `echo`: whether or not to display code in knitted output
- `eval`: whether or to run the code in the chunk when knitting
- `include`: whether to include anything from the from a code chunk in the output document
- `fig.cap`: figure caption
- `fig.scap`: short figure caption, which will be used in the ‘List of Figures’ in the PDF front matter

IMPORTANT: Do *not* use underscoores in your chunk labels - if you do, you are likely to get an error in PDF output saying something like “! Package caption Error: \caption outside float”.

2.2.1. Setup chunks - setup, images, plots

An R Markdown document usually begins with a chunk that is used to **load libraries**, and to **set default chunk options** with `knitr::opts_chunk$set`.

In your thesis, this will probably happen in **index.Rmd** and/or as opening chunks in each of your chapters.

```
```${r setup, include=FALSE}
don't show code unless we explicitly set echo = TRUE
knitr::opts_chunk$set(echo = FALSE)

library(tidyverse)
```
```

2.2.2. Including images

Code chunks are also used for including images, with `include_graphics` from the `knitr` package, as in [Figure 2.2](#)

```
knitr::include_graphics("figures/sample-content/beltcrest.png")
```

Useful chunk options for figures include:



Figura 2.2: Logotipo institucional (Universidad de las Fuerzas Armadas ESPE)

- `out.width` (use with a percentage) for setting the image size
- if you've got an image that gets waaay to big in your output, it will be constrained to the page width by setting `out.width = "100%"`

Figure rotation

You can use the chunk option `out.extra` to rotate images.

The syntax is different for LaTeX and HTML, so for ease we might start by assigning the right string to a variable that depends on the format you're outputting to:

```
if (knitr::is_latex_output()){  
  rotate180 <- "angle=180"  
} else {  
  rotate180 <- "style='transform:rotate(180deg);'"  
}
```

Then you can reference that variable as the value of `out.extra` to rotate images, as in Figure 2.3.

2.2.3. Including plots

Similarly, code chunks are used for including dynamically generated plots. You use ordinary code in R or other languages - Figure 2.4 shows a plot of the `cars` dataset of stopping distances for cars at various speeds (this dataset is built in to **R**).



Figura 2.3: Logotipo institucional con rotación

```
cars %>%  
  ggplot() +  
    aes(x = speed, y = dist) +  
    geom_point()
```

Under the hood, plots are included in your document in the same way as images - when you build the book or knit a chapter, the plot is automatically generated from your code, saved as an image, then included into the output document.

2.2.4. Including tables

Tables are usually included with the `kable` function from the `knitr` package.

Table 2.1 shows the first rows of that cars data - read in your own data, then use this approach to automatically generate tables.

```
cars %>%  
  head() %>%  
  knitr::kable(caption = "A knitr kable table")
```

- Gotcha: when using `kable`, captions are set inside the `kable` function
- The `kable` package is often used with the `kableExtra` package

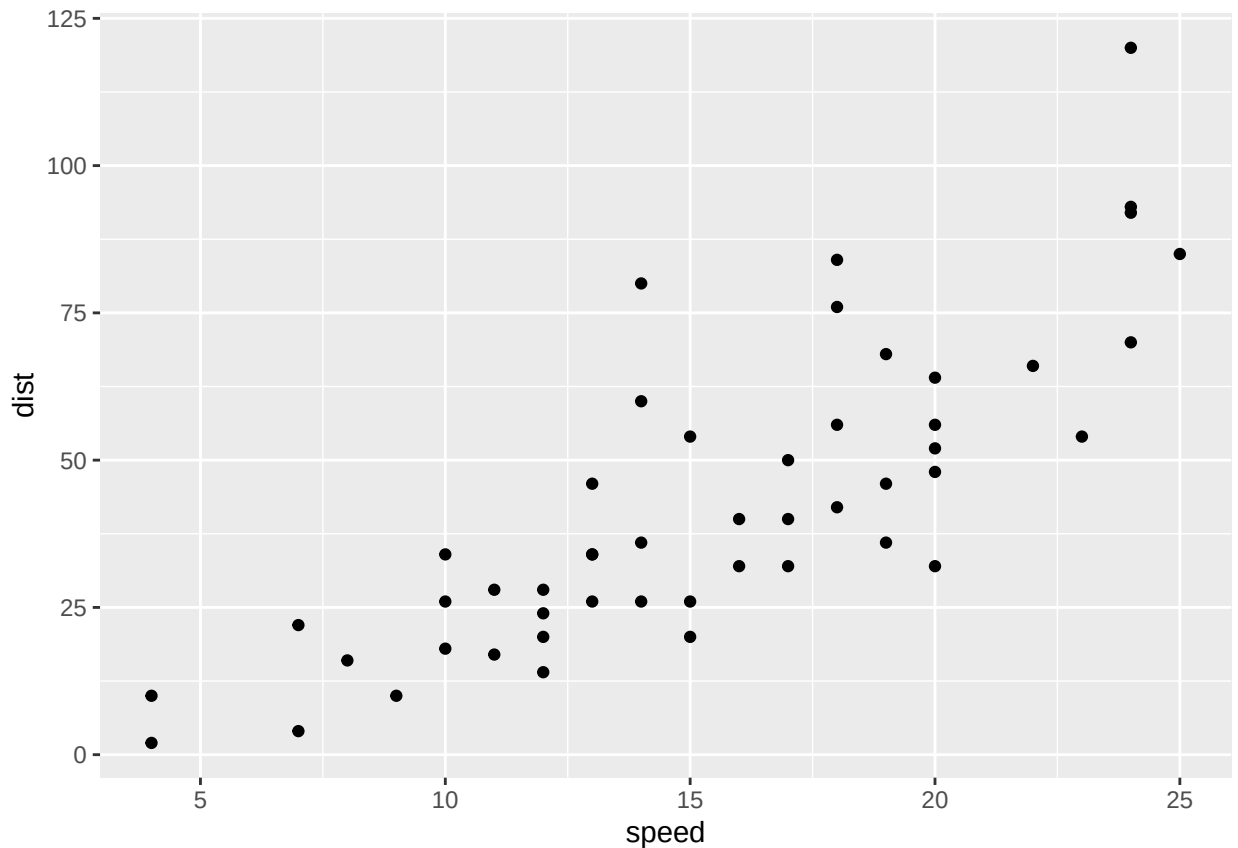


Figura 2.4: A ggplot of car stuff

Tabla 2.1: A knitr kable table

| speed | dist |
|-------|------|
| 4 | 2 |
| 4 | 10 |
| 7 | 4 |
| 7 | 22 |
| 8 | 16 |
| 9 | 10 |

2.2.5. Control positioning

One thing that may be annoying is the way *R Markdown* handles “floats” like tables and figures. In your PDF output, LaTeX will try to find the best place to put your object based on the text around it and until you’re really, truly done writing you should just leave it where it lies.

In general, you should allow LaTeX to do this, but if you really *really* need a

figure to be positioned where you put in the document, then you can make LaTeX attempt to do this with the chunk option `fig.pos="H"`, as in Figure 2.5:

```
knitr::include_graphics("figures/sample-content/beltcrest.png")
```



Figura 2.5: Logotipo institucional posicionado en el flujo de texto

As anyone who has tried to manually play around with the placement of figures in a Word document knows, this can have lots of side effects with extra spacing on other pages, etc. Therefore, it is not generally a good idea to do this - only do it when you really need to ensure that an image follows directly under text where you refer to it (in this document, I needed to do this for Figure 4.1 in section 4.1.4). For more details, read the relevant section of the [R Markdown Cookbook](#).

2.3. Executable inline code

‘Inline code’ simply means inclusion of code inside text. The syntax for doing this is ``r R_CODE``. For example, ``r 4 + 4`` will output 8 in your text.

You will usually use this in parts of your thesis where you report results - read in data or results in a code chunk, store things you want to report in a variable, then insert the value of that variable in your text. For example, we might assign the number of rows in the `cars` dataset to a variable:

2. *R Markdown basics*

```
num_car_observations <- nrow(cars)
```

We might then write:

“In the `cars` dataset, we have ``r num_car_observations`` observations.”

Which would output:

“In the `cars` dataset, we have 50 observations.”

2.4. Executable code in other languages than R

If you want to use other languages than R, such as Python, Julia C++, or SQL, see [the relevant section of the *R Markdown Cookbook*](#)

3

Citations, cross-references, and collaboration

Contents

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3.1. Citations

The usual way to include citations in an *R Markdown* document is to put references in a plain text file with the extension **.bib**, in **BibTeX** format.¹ Then reference the path to this file in **index.Rmd**'s YAML header with **bibliography**: **example.bib**.

¹The bibliography can be in other formats as well, including EndNote (**.enl**) and RIS (**.ris**), see rmarkdown.rstudio.com/authoring_bibliographies_and_citations.

3. Citations and cross-refs

Most reference managers can create a .bib file with your references automatically. However, the **by far** best reference manager to use with *R Markdown* is [Zotero](#) with the [Better BibTex plug-in](#), because the `citr` plugin for RStudio (see below) can read references directly from your Zotero library!

Here is an example of an entry in a **.bib** file:

```
@article{Shea2014,  
  author = {Shea, Nicholas and Boldt, Annika},  
  journal = {Trends in Cognitive Sciences},  
  pages = {186--193},  
  title = {{Supra-personal cognitive control}},  
  volume = {18},  
  year = {2014},  
  doi = {10.1016/j.tics.2014.01.006},  
}
```

In this entry highlighted section, ‘Shea2014’ is the **citation identifier**. To default way to cite an entry in your text is with this syntax: `[@citation-identifier]`.

So I might cite some things (Lottridge et al., 2012; Mill, 1965 [1843]; Shea et al., 2014).

3.1.1. Appearance of citations and references section (pandoc)

By default, this template lets [Pandoc](#) handle how citations are inserted in your text and the references section. You can change the appearance of citations and references by specifying a CSL (Citation Style Language) file in the `cs1` metadata field of **index.Rmd**. By default, citation formatting follows the American Psychological Association (APA 7th Edition), which is an author-year format.

With this style, a number of variations on the citation syntax are useful to know:

- Put author names outside the parenthesis
 - This: `@Shea2014` says blah.
 - Becomes: Shea et al. (2014) says blah.
- Include only the citation-year (in parenthesis)

3. Citations and cross-refs

- This: `Shea et al. says blah [-@Shea2014]`
- Becomes: Shea et al. says blah (2014)
- Add text and page or chapter references to the citation
 - This: `[see @Shea2014, pp. 33-35; also @Wu2016, ch. 1]`
 - Becomes: Blah blah (see Shea et al., 2014, pp. 33-35; also Wu, 2016, ch. 1).

If you want a numerical citation style instead, try `csl: bibliography/transactions-on-comput` or just have a browse through the [Zotero Style Repository](#) and look for one you like. For convenience, you can set the line spacing and the space between the bibliographic entries in the reference section directly from the YAML header in **index.Rmd**.

If you prefer to use `biblatex` or `natbib` to handle references, see [this chapter](#).

3. Citations and cross-refs

3.1.2. Insert references easily with RStudio's Visual Editor

For an easy way to insert citations, use RStudio's [Visual Editor](#). Make sure you have the latest version of RStudio – the visual editor was originally really buggy, especially in relation to references, but as per v2022.02.0, it's great!

3.2. Cross-referencing

We can make cross-references to **sections** within our document, as well as to **figures** (images and plots) and **tables**.

The general cross-referencing syntax is `\@ref(label)`

3.2.1. Section references

Headers are automatically assigned a reference label, which is the text in lower caps separated by dashes. For example, `# My header` is automatically given the label `my-header`. So `# My header` can be referenced with `\@ref(my-section)`

Remember what we wrote in [section 3.1](#)?

We can also use **hyperlink syntax** and add `#` before the label, though this is only guaranteed to work properly in HTML output:

- So if we write `Remember what we wrote up in [the previous section](#citations)?`
- It becomes `Remember what we wrote up in the previous section?`

Creating custom labels

It is a very good idea to create **custom labels** for our sections. This is because the automatically assigned labels will change when we change the titles of the sections - to avoid this, we can create the labels ourselves and leave them untouched if we change the section titles.

We create custom labels by adding `{#label}` after a header, e.g. `# My section {#my-label}`. See [our chapter title](#) for an example. That was section 3.

3. Citations and cross-refs



Figura 3.1: A marvel-lous meme

3.2.2. Figure (image and plot) references

- To refer to figures (i.e. images and plots) use the syntax `\@ref(fig:label)`
- **GOTCHA:** Figures and tables must have captions if you wish to cross-reference them.

Let's add an image:

```
knitr::include_graphics("figures/sample-content/captain.jpeg")
```

We refer to this image with `\@ref(fig:captain)`. So Figure 3.1 is [this image](#).

And in Figure 2.4 we saw a [cars](#) plot.

3.2.3. Table references

- To refer to tables use the syntax `\@ref(tab:label)`

Let's include a table:

3. Citations and cross-refs

Table 3.1: Stopping cars

| speed | dist |
|-------|------|
| 4 | 2 |
| 4 | 10 |
| 7 | 4 |
| 7 | 22 |
| 8 | 16 |

```
knitr::kable(cars[1:5,],  
             caption="Stopping cars")
```

We refer to this table with `\@ref(tab:cars-table2)`. So Table 3.1 is [this table](#).

And in Table 2.1 we saw more or less [the same cars table](#).

3.2.4. Including page numbers

Finally, in the PDF output we might also want to include the page number of a reference, so that it's easy to find in physical printed output. LaTeX has a command for this, which looks like this: `\pageref{fig/tab:label}` (note: curly braces, not parentheses)

When we output to PDF, we can use raw LaTeX directly in our .Rmd files. So if we wanted to include the page of the cars plot we could write:

- This: Figure `\@ref(fig:cars-plot)` on page `\pageref(fig:cars-plot)`
- Becomes: Figure [2.4](#) on page [17](#)

Include page numbers only in PDF output

A problem here is that LaTeX commands don't display in HTML output, so in the gitbook output we'd see simply "Figure [2.4](#) on page".

One way to get around this is to use inline R code to insert the text, and use an `ifelse` statement to check the output format and then insert the appropriate text.

- So this: ``r ifelse(knitr::is_latex_output(), "Figure \@ref(fig:cars-plot) on page \pageref{fig:cars-plot}", )``

3. Citations and cross-refs

- Inserts this (check this on both PDF and gitbook): Figure 2.4 on page 17

Note that we need to escape the backslash with another backslash here to get the correct output.

3.3. Collaborative writing

Best practices for collaboration and change tracking when using R Markdown are still an open question. In the blog post [One year to dissertate](#) by Lucy D’Agostino, which I highly recommend, the author notes that she knits .Rmd files to a word document, then uses the `googledrive` R package to send this to Google Drive for comments / revisions from co-authors, then incorporates Google Drive suggestions *by hand* into the .Rmd source files. This is a bit clunky, and there are ongoing discussions among the *R Markdown* developers about what the best way is to handle collaborative writing (see [issue #1463](#) on GitHub, where [CriticMarkup](#) is among the suggestions).

For now, this is an open question in the community of R Markdown users. I often knit to a format that can easily be imported to Google Docs for comments, then go over suggested revisions and manually incorporate them back in to the .Rmd source files. For articles, I sometimes upload a near-final draft to [Overleaf](#), then collaboratively make final edits to the LaTeX file there. I suspect some great solution will be developed in the not-to-distant future, probably by the RStudio team.

3.4. Additional resources

- *R Markdown: The Definitive Guide* - <https://bookdown.org/yihui/rmarkdown/>
- *R for Data Science* - <https://r4ds.had.co.nz>

4

Tables

Contents

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4.1. Making LaTeX tables play nice

Dealing with tables in LaTeX can be painful. This section explains the main tricks you need to make the pain go away.

(Note: if you are looking at the ebook version, you will not see much difference in this section, as it is only relevant for PDF output!)

4.1.1. Making your table pretty

When you use `kable` to create tables, you will almost certainly want to set the option `booktabs = TRUE`. This makes your table look a million times better:

4. Tables

```
library(knitr)
library(tidyverse)

head(mtcars) %>%
  kable(booktabs = TRUE)
```

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|-------------------|------|-----|------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4 | 21.0 | 6 | 160 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag | 21.0 | 6 | 160 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 710 | 22.8 | 4 | 108 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive | 21.4 | 6 | 258 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout | 18.7 | 8 | 360 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant | 18.1 | 6 | 225 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |

Compare this to the default style, which looks terrible:

```
head(mtcars) %>%
  kable()
```

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|-------------------|------|-----|------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4 | 21.0 | 6 | 160 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag | 21.0 | 6 | 160 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 710 | 22.8 | 4 | 108 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive | 21.4 | 6 | 258 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout | 18.7 | 8 | 360 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant | 18.1 | 6 | 225 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |

4.1.2. If your table is too wide

You might find that your table expands into the margins of the page, like the tables above. Fix this with the `kable_styling` function from the `kableExtra` package:

```
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.4.3
```

```
head(mtcars) %>%
  kable(booktabs = TRUE) %>%
  kable_styling(latex_options = "scale_down")
```

This scales down the table to fit the page width.

4. Tables

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|-------------------|------|-----|------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4 | 21.0 | 6 | 160 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag | 21.0 | 6 | 160 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 710 | 22.8 | 4 | 108 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive | 21.4 | 6 | 258 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout | 18.7 | 8 | 360 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant | 18.1 | 6 | 225 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |

4.1.3. If your table is too long

If your table is too long to fit on a single page, set `longtable = TRUE` in the `kable` function to split the table across multiple pages.

```
a_long_table <- rbind(mtcars, mtcars)

a_long_table %>%
  select(1:8) %>%
  kable(booktabs = TRUE, longtable = TRUE)
```

| | mpg | cyl | disp | hp | drat | wt | qsec | vs |
|---------------------|------|-----|-------|-----|------|-------|-------|----|
| Mazda RX4 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 |
| Mazda RX4 Wag | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 |
| Datsun 710 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 |
| Hornet 4 Drive | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 |
| Hornet Sportabout | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 |
| Valiant | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 |
| Duster 360 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 |
| Merc 240D | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 |
| Merc 230 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 |
| Merc 280 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 |
| Merc 280C | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 |
| Merc 450SE | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 |
| Merc 450SL | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 |
| Merc 450SLC | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 |
| Cadillac Fleetwood | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 |
| Lincoln Continental | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 |
| Chrysler Imperial | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 |
| Fiat 128 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 |
| Honda Civic | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 |
| Toyota Corolla | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 |
| Toyota Corona | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 |

4. Tables

| | | | | | | | | |
|----------------------|------|---|-------|-----|------|-------|-------|---|
| Dodge Challenger | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 |
| AMC Javelin | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 |
| Camaro Z28 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 |
| Pontiac Firebird | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 |
| Fiat X1-9 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 |
| Porsche 914-2 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 |
| Lotus Europa | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 |
| Ford Pantera L | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 |
| Ferrari Dino | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 |
| Maserati Bora | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 |
| Volvo 142E | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 |
| Mazda RX41 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 |
| Mazda RX4 Wag1 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 |
| Datsun 7101 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 |
| Hornet 4 Drive1 | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 |
| Hornet Sportabout1 | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 |
| Valiant1 | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 |
| Duster 3601 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 |
| Merc 240D1 | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 |
| Merc 2301 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 |
| Merc 2801 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 |
| Merc 280C1 | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 |
| Merc 450SE1 | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 |
| Merc 450SL1 | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 |
| Merc 450SLC1 | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 |
| Cadillac Fleetwood1 | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 |
| Lincoln Continental1 | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 |
| Chrysler Imperial1 | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 |
| Fiat 1281 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 |
| Honda Civic1 | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 |
| Toyota Corolla1 | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 |
| Toyota Corona1 | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 |
| Dodge Challenger1 | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 |
| AMC Javelin1 | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 |
| Camaro Z281 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 |
| Pontiac Firebird1 | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 |
| Fiat X1-91 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 |
| Porsche 914-21 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 |
| Lotus Europa1 | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 |
| Ford Pantera L1 | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 |
| Ferrari Dino1 | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 |
| Maserati Bora1 | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 |

4. Tables

| | | | | | | | | |
|-------------|------|---|-------|-----|------|-------|-------|---|
| Volvo 142E1 | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 |
|-------------|------|---|-------|-----|------|-------|-------|---|

When you do this, you'll probably want to make the header repeat on new pages. Do this with the `kable_styling` function from `kableExtra`:

```
a_long_table %>%
  kable(booktabs = TRUE, longtable = TRUE) %>%
  kable_styling(latex_options = "repeat_header")
```

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|---------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 710 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |
| Duster 360 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 | 0 | 3 | 4 |
| Merc 240D | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 | 0 | 4 | 2 |
| Merc 230 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 | 0 | 4 | 2 |
| Merc 280 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 | 0 | 4 | 4 |
| Merc 280C | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 | 0 | 4 | 4 |
| Merc 450SE | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 | 0 | 3 | 3 |
| Merc 450SL | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 | 3 |
| Merc 450SLC | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 | 3 |
| Cadillac Fleetwood | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 | 4 |
| Lincoln Continental | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 | 4 |
| Chrysler Imperial | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 | 4 |
| Fiat 128 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 | 1 |
| Honda Civic | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 | 2 |
| Toyota Corolla | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 | 1 |
| Toyota Corona | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 | 1 |
| Dodge Challenger | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 | 2 |
| AMC Javelin | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 | 2 |
| Camaro Z28 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 | 4 |
| Pontiac Firebird | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 | 2 |
| Fiat X1-9 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 | 1 |
| Porsche 914-2 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 | 2 |
| Lotus Europa | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 | 2 |
| Ford Pantera L | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 | 4 |
| Ferrari Dino | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 | 6 |
| Maserati Bora | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 | 8 |

4. Tables

(continued)

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|----------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Volvo 142E | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 | 2 |
| Mazda RX41 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag1 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 7101 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive1 | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout1 | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant1 | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |
| Duster 3601 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 | 0 | 3 | 4 |
| Merc 240D1 | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 | 0 | 4 | 2 |
| Merc 2301 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 | 0 | 4 | 2 |
| Merc 2801 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 | 0 | 4 | 4 |
| Merc 280C1 | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 | 0 | 4 | 4 |
| Merc 450SE1 | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 | 0 | 3 | 3 |
| Merc 450SL1 | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 | 3 |
| Merc 450SLC1 | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 | 3 |
| Cadillac Fleetwood1 | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 | 4 |
| Lincoln Continental1 | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 | 4 |
| Chrysler Imperial1 | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 | 4 |
| Fiat 1281 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 | 1 |
| Honda Civic1 | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 | 2 |
| Toyota Corolla1 | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 | 1 |
| Toyota Corona1 | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 | 1 |
| Dodge Challenger1 | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 | 2 |
| AMC Javelin1 | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 | 2 |
| Camaro Z281 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 | 4 |
| Pontiac Firebird1 | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 | 2 |
| Fiat X1-91 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 | 1 |
| Porsche 914-21 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 | 2 |
| Lotus Europa1 | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 | 2 |
| Ford Pantera L1 | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 | 4 |
| Ferrari Dino1 | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 | 6 |
| Maserati Bora1 | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 | 8 |
| Volvo 142E1 | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 | 2 |

Unfortunately, we cannot use the `scale_down` option with a `longtable`. So if a `longtable` is too wide, you can either manually adjust the font size, or show the table in landscape layout. To adjust the font size, use `kableExtra`'s `font_size` option:

4. Tables

```
a_long_table %>%
  kable(booktabs = TRUE, longtable = TRUE) %>%
  kable_styling(font_size = 9, latex_options = "repeat_header")
```

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|---------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 710 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |
| Duster 360 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 | 0 | 3 | 4 |
| Merc 240D | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 | 0 | 4 | 2 |
| Merc 230 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 | 0 | 4 | 2 |
| Merc 280 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 | 0 | 4 | 4 |
| Merc 280C | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 | 0 | 4 | 4 |
| Merc 450SE | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 | 0 | 3 | 3 |
| Merc 450SL | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 | 3 |
| Merc 450SLC | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 | 3 |
| Cadillac Fleetwood | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 | 4 |
| Lincoln Continental | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 | 4 |
| Chrysler Imperial | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 | 4 |
| Fiat 128 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 | 1 |
| Honda Civic | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 | 2 |
| Toyota Corolla | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 | 1 |
| Toyota Corona | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 | 1 |
| Dodge Challenger | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 | 2 |
| AMC Javelin | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 | 2 |
| Camaro Z28 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 | 4 |
| Pontiac Firebird | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 | 2 |
| Fiat X1-9 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 | 1 |
| Porsche 914-2 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 | 2 |
| Lotus Europa | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 | 2 |
| Ford Pantera L | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 | 4 |
| Ferrari Dino | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 | 6 |
| Maserati Bora | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 | 8 |
| Volvo 142E | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 | 2 |
| Mazda RX41 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag1 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 7101 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive1 | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout1 | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant1 | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |
| Duster 3601 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 | 0 | 3 | 4 |
| Merc 240D1 | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 | 0 | 4 | 2 |
| Merc 2301 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 | 0 | 4 | 2 |
| Merc 2801 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 | 0 | 4 | 4 |
| Merc 280C1 | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 | 0 | 4 | 4 |
| Merc 450SE1 | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 | 0 | 3 | 3 |
| Merc 450SL1 | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 | 3 |
| Merc 450SLC1 | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 | 3 |
| Cadillac Fleetwood1 | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 | 4 |

4. Tables

(continued)

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|----------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Lincoln Continental1 | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 | 4 |
| Chrysler Imperial1 | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 | 4 |
| Fiat 1281 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 | 1 |
| Honda Civic1 | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 | 2 |
| Toyota Corolla1 | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 | 1 |
| Toyota Corona1 | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 | 1 |
| Dodge Challenger1 | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 | 2 |
| AMC Javelin1 | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 | 2 |
| Camaro Z281 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 | 4 |
| Pontiac Firebird1 | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 | 2 |
| Fiat X1-91 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 | 1 |
| Porsche 914-21 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 | 2 |
| Lotus Europa1 | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 | 2 |
| Ford Pantera L1 | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 | 4 |
| Ferrari Dino1 | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 | 6 |
| Maserati Bora1 | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 | 8 |
| Volvo 142E1 | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 | 2 |

To put the table in landscape mode, use kableExtra's `landscape` function:

```
a_long_table %>%
  kable(booktabs = TRUE, longtable = TRUE) %>%
  kable_styling(latex_options = "repeat_header") %>%
  landscape()
```

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|---------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 710 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |
| Duster 360 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 | 0 | 3 | 4 |
| Merc 240D | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 | 0 | 4 | 2 |
| Merc 230 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 | 0 | 4 | 2 |
| Merc 280 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 | 0 | 4 | 4 |
| Merc 280C | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 | 0 | 4 | 4 |
| Merc 450SE | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 | 0 | 3 | 3 |
| Merc 450SL | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 | 3 |
| Merc 450SLC | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 | 3 |
| Cadillac Fleetwood | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 | 4 |
| Lincoln Continental | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 | 4 |
| Chrysler Imperial | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 | 4 |
| Fiat 128 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 | 1 |
| Honda Civic | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 | 2 |
| Toyota Corolla | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 | 1 |
| Toyota Corona | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 | 1 |
| Dodge Challenger | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 | 2 |
| AMC Javelin | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 | 2 |
| Camaro Z28 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 | 4 |
| Pontiac Firebird | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 | 2 |

(continued)

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|----------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Fiat X1-9 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 | 1 |
| Porsche 914-2 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 | 2 |
| Lotus Europa | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 | 2 |
| Ford Pantera L | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 | 4 |
| Ferrari Dino | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 | 6 |
| Maserati Bora | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 | 8 |
| Volvo 142E | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 | 2 |
| Mazda RX41 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag1 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 7101 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive1 | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout1 | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant1 | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |
| Duster 3601 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 | 0 | 3 | 4 |
| Merc 240D1 | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 | 0 | 4 | 2 |
| Merc 2301 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 | 0 | 4 | 2 |
| Merc 2801 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 | 0 | 4 | 4 |
| Merc 280C1 | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 | 0 | 4 | 4 |
| Merc 450SE1 | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 | 0 | 3 | 3 |
| Merc 450SL1 | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 | 3 |
| Merc 450SLC1 | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 | 3 |
| Cadillac Fleetwood1 | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 | 4 |
| Lincoln Continental1 | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 | 4 |
| Chrysler Imperial1 | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 | 4 |

(continued)

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|-------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Fiat 1281 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 | 1 |
| Honda Civic1 | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 | 2 |
| Toyota Corolla1 | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 | 1 |
| Toyota Corona1 | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 | 1 |
| Dodge Challenger1 | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 | 2 |
| AMC Javelin1 | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 | 2 |
| Camaro Z281 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 | 4 |
| Pontiac Firebird1 | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 | 2 |
| Fiat X1-91 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 | 1 |
| Porsche 914-21 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 | 2 |
| Lotus Europa1 | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 | 2 |
| Ford Pantera L1 | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 | 4 |
| Ferrari Dino1 | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 | 6 |
| Maserati Bora1 | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 | 8 |
| Volvo 142E1 | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 | 2 |

4. Tables

4.1.4. Max power: manually adjust the raw LaTeX output

For total flexibility, you can adjust the raw LaTeX output from `kable/kableExtra` that generates the table. Let us consider how we would do this for the example of adjusting the font size if our table is too wide: Latex has a bunch of standard commands that set an approximate font size, as shown below in Figure 4.1.

| | |
|----------------------------|-------------|
| <code>\tiny</code> | Lorem ipsum |
| <code>\scriptsize</code> | Lorem ipsum |
| <code>\footnotesize</code> | Lorem ipsum |
| <code>\small</code> | Lorem ipsum |

Figura 4.1: Font sizes in LaTeX

You could use these to manually adjust the font size in your longtable in two steps:

1. Wrap the longtable environment in, e.g., a `scriptsize` environment, by doing a string replacement in the output from `kable/kableExtra`
2. Add the attributes that make R Markdown understand that the table is a table (it seems R drops these when we do the string replacement)

```
our_adjusted_table <- a_long_table %>%
  kable(booktabs = TRUE, longtable = TRUE) %>%
  kable_styling(latex_options = "repeat_header") %>%
  # wrap the longtable in a tiny environment
  str_replace('\\\\begin\\{longtable\\}',
              '\\\\begin\\{scriptsize\\}\\n\\\\begin\\{longtable\\}'
              ↪ '}') %>%
  str_replace('\\\\end\\{longtable\\}',
              '\\\\end\\{longtable\\}\\n\\\\end\\{scriptsize\\}')

#add attributes to make R Markdown treat this as a kable LaTeX
↪ table again
```

4. Tables

```
our_adjusted_table %>%
  structure(format = "latex", class = "knitr_kable")
```

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|----------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 710 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |
| Duster 360 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 | 0 | 3 | 4 |
| Merc 240D | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 | 0 | 4 | 2 |
| Merc 230 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 | 0 | 4 | 2 |
| Merc 280 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 | 0 | 4 | 4 |
| Merc 280C | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 | 0 | 4 | 4 |
| Merc 450SE | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 | 0 | 3 | 3 |
| Merc 450SL | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 | 3 |
| Merc 450SLC | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 | 3 |
| Cadillac Fleetwood | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 | 4 |
| Lincoln Continental | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 | 4 |
| Chrysler Imperial | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 | 4 |
| Fiat 128 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 | 1 |
| Honda Civic | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 | 2 |
| Toyota Corolla | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 | 1 |
| Toyota Corona | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 | 1 |
| Dodge Challenger | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 | 2 |
| AMC Javelin | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 | 2 |
| Camaro Z28 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 | 4 |
| Pontiac Firebird | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 | 2 |
| Fiat X1-9 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 | 1 |
| Porsche 914-2 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 | 2 |
| Lotus Europa | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 | 2 |
| Ford Pantera L | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 | 4 |
| Ferrari Dino | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 | 6 |
| Maserati Bora | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 | 8 |
| Volvo 142E | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 | 2 |
| Mazda RX41 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0 | 1 | 4 | 4 |
| Mazda RX4 Wag1 | 21.0 | 6 | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0 | 1 | 4 | 4 |
| Datsun 7101 | 22.8 | 4 | 108.0 | 93 | 3.85 | 2.320 | 18.61 | 1 | 1 | 4 | 1 |
| Hornet 4 Drive1 | 21.4 | 6 | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1 | 0 | 3 | 1 |
| Hornet Sportabout1 | 18.7 | 8 | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0 | 0 | 3 | 2 |
| Valiant1 | 18.1 | 6 | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1 | 0 | 3 | 1 |
| Duster 3601 | 14.3 | 8 | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0 | 0 | 3 | 4 |
| Merc 240D1 | 24.4 | 4 | 146.7 | 62 | 3.69 | 3.190 | 20.00 | 1 | 0 | 4 | 2 |
| Merc 2301 | 22.8 | 4 | 140.8 | 95 | 3.92 | 3.150 | 22.90 | 1 | 0 | 4 | 2 |
| Merc 2801 | 19.2 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1 | 0 | 4 | 4 |
| Merc 280C1 | 17.8 | 6 | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1 | 0 | 4 | 4 |
| Merc 450SE1 | 16.4 | 8 | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0 | 0 | 3 | 3 |
| Merc 450SL1 | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 | 3 |
| Merc 450SLC1 | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 | 3 |
| Cadillac Fleetwood1 | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 | 4 |
| Lincoln Continental1 | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 | 4 |
| Chrysler Imperial1 | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 | 4 |
| Fiat 1281 | 32.4 | 4 | 78.7 | 66 | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 | 1 |
| Honda Civic1 | 30.4 | 4 | 75.7 | 52 | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 | 2 |
| Toyota Corolla1 | 33.9 | 4 | 71.1 | 65 | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 | 1 |
| Toyota Corona1 | 21.5 | 4 | 120.1 | 97 | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 | 1 |
| Dodge Challenger1 | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 | 2 |
| AMC Javelin1 | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 | 2 |

4. Tables

(continued)

| | mpg | cyl | disp | hp | drat | wt | qsec | vs | am | gear | carb |
|-------------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Camaro Z281 | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 | 4 |
| Pontiac Firebird1 | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 | 2 |
| Fiat X1-91 | 27.3 | 4 | 79.0 | 66 | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 | 1 |
| Porsche 914-21 | 26.0 | 4 | 120.3 | 91 | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 | 2 |
| Lotus Europa1 | 30.4 | 4 | 95.1 | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 | 2 |
| Ford Pantera L1 | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 | 4 |
| Ferrari Dino1 | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 | 6 |
| Maserati Bora1 | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 | 8 |
| Volvo 142E1 | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 | 2 |

There is grandeur in this view of life, with its several powers, having been originally breathed into a few forms or into one; and that, whilst this planet has gone cycling on according to the fixed law of gravity, from so simple a beginning endless forms most beautiful and most wonderful have been, and are being, evolved.

— Charles Darwin (Darwin, 1859)

5

Customisations and extensions

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Este capítulo describe recomendaciones, trucos y posibles personalizaciones para enriquecer este trabajo de investigación.

5.1. Chunk caching and the `__bookdown__files` folder

If you set `cache=TRUE` in a code chunk, in order to cache its results if it's time-consuming to run see [the R Markdown documentation](#), then the files for the caching are stored in the `**__bookdown__files**` folder.

If you don't use caching and you would like to just have the `**__bookdown__files**` folder deleted after the build process is complete, then set `allow_cache = FALSE` in `index.Rmd`'s call to `knit_thesis`.

That is, your YAML should then look like this:

```
knit: (function(input, ...) {  
  thesis_formats <- "pdf";  
  
  source("scripts_and_filters/knit-functions.R");  
  knit_thesis(input, thesis_formats, allow_cache = FALSE, ...)  
})
```

5.2. Front matter

5.2.1. Shorten captions shown in the list of figures (PDF)

You might want your list of figures (which follows the table of contents) to have shorter (or just different) figure descriptions than the actual figure captions.

Do this using the chunk option `fig.scap` ('short caption'), for example `{r captain-image, fig.cap=.^ very long and descriptive (and potentially boring)`

5. Customisations and extensions

caption that doesn't fit in the list of figures, but helps the reader understand what the figure communicates.", fig.scap=.^A concise description for the list of figures"

5.2.2. Shorten captions shown in the list of tables (PDF)

You might want your list of tables (which follows the list of figures in your thesis front matter) to have shorter (or just different) table descriptions than the actual table captions.

If you are using `knitr::kable` to generate a table, you can do this with the argument `caption.short`, e.g.:

```
knitr::kable(mtcars,  
             caption = "A very long and descriptive (and  
             ↪ potentially  
             boring) caption that doesn't fit in the list of  
             ↪ figures,  
             but helps the reader understand what the figure  
             communicates.",  
             caption.short = "A concise description for the list of  
             ↪ tables")
```

5.3. Shorten running header (PDF)

You might want a chapter's running header (i.e. the header showing the title of the current chapter at the top of page) to be shorter (or just different) to the actual chapter title.

Do this by adding the latex command `\chaptermark{My shorter version}` after your chapter title.

For example, chapter 3's running header is simply 'Cites and cross-refs', because it begins like this:

```
# Citations, cross-references, and collaboration {#cites-and-refs}  
↪  
\chaptermark{Cites and cross-refs}
```

5.4. Unnumbered chapters

To make chapters unnumbered (normally only relevant to the Introduction and/or the Conclusion), follow the chapter header with `{-}`, e.g. `# Introduction {-}`.

When you do this, you must also follow the heading with these two latex commands:

```
\adjustmtc
\markboth{The Name of Your Unnumbered Chapter}{{}}
```

Otherwise the chapter's mini table of contents and the running header will show the previous chapter.

5.5. Beginning chapters with quotes (PDF)

The OxThesis LaTeX template lets you inject some wittiness into your thesis by including a block of type `savequote` at the beginning of chapters. To do this, use the syntax ````{block type='savequote'}`.¹

Add the reference for the quote with the chunk option `quote_author="my author name"`. You will also want to add the chunk option `include=knitr::is_latex_output()` so that quotes are only included in PDF output.

It's not possible to use markdown syntax inside chunk options, so if you want to e.g. italicise a book name in the reference use a `'text reference'`: Create a named piece of text with `'(ref:label-name) My text'`, then point to this in the chunk option with `quote_author='(ref:label-name)'`.

5.6. Highlighting corrections (HTML & PDF)

For when it comes time to do corrections, you may want to highlight changes made when you submit a post-viva, corrected copy to your examiners so they can

¹For more on custom block types, see the relevant section in *Authoring Books with R Markdown*.

quickly verify you've completed the task. You can do so like this:

5.6.1. Short, inline corrections

Highlight **short, inline corrections** by doing `[like this]{.correction}` — the text between the square brackets will then **be highlighted in blue** in the output.

Note that pandoc and LaTeX get confused by citations and cross-references inside inline corrections. In particular, avoid doing `"[what @Shea2014 said]{.correction}"` directly in inline text, as the underlying LaTeX package (`soul`) cannot parse citations inside `\hl{}`. In such cases, use an explicit correction block or write the citation outside the highlighted text:

- For highlighting text next to a citation: `[what was said]{.correction} by @Shea2014`
- Or use a correction block ````{block type='correction'}` for larger segments.

5.6.2. Blocks of added or changed material

Highlight entire **blocks of added or changed material** by putting them in a block of type `correction`, using the syntax ````{block type='correction'}`.² Like so:

For larger chunks, like this paragraph or indeed entire figures, you can use the `correction` block type. This environment **highlights paragraph-sized and larger blocks** with the same blue colour.

Note that correction blocks cannot be included in word output.

5.6.3. Stopping corrections from being highlighted

To turn off correction highlighting, go to the YAML header of `index.Rmd`, then:

²In the `.tex` file for PDF output, this will put the content between `\begin{correction}` and `\end{correction}`; in gitbook output it will be put between `<div class=correction>` and `</div>`.

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- PDF output: set `corrections: false`
- HTML output: remove or comment out - `templates/corrections.css`

5.7. Apply custom font color and highlighting to text (HTML & PDF)

The lua filter that adds the functionality to highlight corrections adds two more tricks: you can apply your own choice of colour to highlight text, or change the font color. The syntax is as follows:

Here's [some text in pink highlighting]{highlight="pink"}
Becomes: Here's some text in pink highlighting.

[Here's some text with blue font]{color="blue"}
Becomes: Here's some text with blue font

Finally — never, ever actually do this - [here's some text with black highlighting and yellow font]{highlight="black" color="yellow"}
Becomes: here's some text with black highlighting and yellow font

The file `scripts_and_filters/colour_and_highlight.lua` implements this, if you want to fiddle around with it. It works with both PDF and HTML output.

5.8. Adding a second abstract (PDF)

You may need two abstracts in your thesis, if you e.g. need both an abstract in English and some other language.

You can add a second abstract in `index.Rmd` like so:

```
abstract-second-heading: "Resumé"
abstract-second: "This is the second abstract, for example in
↪ beautiful French."
```

5.9. Including another paper in your thesis - embed a PDF document

You may want to embed existing PDF documents into the thesis, for example if your department allows a ‘portfolio’ style thesis and you need to include an existing typeset publication as a chapter.

In gitbook output, you can simply use `knitr::include_graphics` and it should include a scrollable (and downloadable) PDF. You will probably want to set the chunk options `out.width='100%'` and `out.height='1000px'`:

```
# Reemplaza la ruta con su propio archivo PDF para incrustar:  
# knitr::include_graphics("figures/mi_documento.pdf")
```

In LaTeX output, however, this approach can cause odd behaviour. Therefore, when you build your thesis to PDF, split the PDF into an alphanumerically sorted sequence of **single-page** PDF files (you can do this automatically with the package `pdftools`). You can then use the appropriate LaTeX command to insert them, as shown below. *Note that the chunk option `results='asis'` must be set.* You may also want to remove margins from the PDF files, which you can do with Adobe Acrobat (paid version) and likely other software.

```
# install.packages(pdftools)  
# Divida su PDF en páginas individuales:  
# pdftools::pdf_split("figures/mi_documento.pdf",  
# output = "figures/mi_documento_split/")  
  
# Tome las páginas  
pages <- list.files("figures/mi_documento_split", full.names =  
  ↪ TRUE)  
  
# Establezca el ancho de los PDFs insertados:  
# 1.0 es el 100% del ancho de la página PDF  
pdf_width <- 1.2  
  
# Inserte cada página del PDF  
cat(stringr::str_c("\\newpage \\begin{center}  
  \\makebox[\\linewidth]{c}{\\includegraphics[width=", pdf_width,  
  "\\linewidth]{", pages, "}} \\end{center}"))
```

5.10. Including another paper in your thesis - R Markdown child document

Sometimes you want to include another paper you are currently writing as a chapter in your thesis. Above 5.9, we described the simplest way to do this: include the other paper as a pdf. However, in some cases you instead want to include the R Markdown source from this paper, and have it compiled within your thesis. This is a little bit more tricky, because you need to keep careful track of your file paths, but it is possible by [including the paper as a child document](#). There are four main steps:

1. Include the paper as a child document
2. Make file paths compatible with knitting the article on its own, as well as when it's include in your thesis
3. Make header levels correct
4. Make figure widths correct

5.10.1. An example paper in another folder

Considere este ejemplo sencillo (los archivos pueden almacenarse en un repositorio externo):

```
/--paper_to_include
/  /--my_paper.Rmd
/  /--data
/  /  /--cat_salt.csv
/  /--figures
/  /  /--cat.jpg
/
/--thesis
```

As the chart suggests, you have another folder, **paper_to_include/** living in the same containing folder as your thesis folder. In the **paper_to_include** folder, the file **my_paper.Rmd** is where you write the paper. In **my_paper.Rmd**, you read in a CSV file found in the subfolder **data/cats.csv**, and also an image from the subfolder **figures/cat.jpg**.

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5.10.2. Step 1: Include paper as a child document

In your thesis folder, create an Rmd file for the chapter where you want to include another paper. Add one or more code chunks that include R Markdown files from that paper as child documents:

```
# Including an external chapter

```{r child = "../paper_to_include/my_paper.Rmd"}
```
```

5.10.3. Step 2: Make file paths compatible

Use [parameters](#) to adjust the file path of images based on values you set in the YAML header of an R Markdown file. In **my_paper.Rmd**, create a parameter called `other_path` and set it to an empty string:

```
---
title: "A fabulous article in a different folder"
params:
  other_path: ""
---
```

In **my_paper.Rmd**, put this at the start of the filepath when you read in data or include images:

```
library(tidyverse)
library(knitr)

cat_data <- read_csv(str_c(params$other_path, "data/cats.csv"))
include_graphics(str_c(params$other_path, "figures/cat.jpg"))
```

Finally, in your thesis folder's **index.Rmd** file, also create the parameter `other_path`. But here, set it to where the **paper_to_include/** folder is relative to your thesis folder:

```
params:
  other_path: "../paper_to_include/"
```

5. Customisations and extensions

Note on HTML output

Note that if you want to host an HTML version on your thesis online, you will need to include graphics in the content that you host online - the internet obviously won't be able to see filepaths that are just referring to stuff in another folder on your computer!

5.10.4. Step 3: Make sure header levels are correct

Unless the paper you want to include is also written as a book, your header levels are probably going to be off. That is, the level 1 headers (`# Some header`) you use for main sections in the other paper turns into chapter titles when included in your thesis.

To avoid this, first *increment all heading levels by one in `paper_to_include/my_paper.Rmd`* (`# Some header -> ## Some header`). Then in `paper_to_include/` create a [lua filter](#) that decrements header levels by one: Create a text file, save it as `reduce_header_level.lua`, and give it the content below.

```
function Header(el)
  if (el.level <= 1) then
    error("I don't know how to decrease the level of h1")
  end
  el.level = el.level - 1
  return el
end
```

In the YAML header of `paper_to_include/my_paper.Rmd`, use this filter:

```
---
title: "A fabulous article in a different folder"
params:
  other_path: ""
output:
  pdf_document:
    pandoc_args: ["--lua-filter=reduce_header_level.lua"]
---
```

Now, your header levels will be correct both when you knit the paper on its own and when its included in your thesis.

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NOTE: There might be no need to use a lua filter to shift heading - it seems you could simply use `pandoc_args: [-shift-heading-level-by=-1]` (see <https://pandoc.org/MANUAL.html#reader-options>)

5.10.5. Step 4. Make sure figure widths are correct

It might be that your figure widths when knitting your paper on its own, and when including it in your thesis, need to be different. You can again use parameters to set figure widths.

Imagine you want figure width to be 80 % of the page width when knitting your paper on its own, but 100 % in your thesis. In `paper_to_include/my_paper.Rmd`, first add a parameter we could call `out_width` and set it to the string "80 %":

```
---
title: "A fabulous article in a different folder"
params:
  other_path: ""
  out_width: "80%"
output:
  pdf_document:
    pandoc_args: ["--lua-filter=reduce_header_level.lua"]
---
```

Then, make sure use that parameter to set the output width when you include figures in `paper_to_include/my_paper.Rmd`:

```
```{r, out.width=params$out_width, fig.cap="A very funny cat"}
include_graphics(str_c(params$other_path, "figures/cat.jpg"))
```
```

Finally, create the parameter `out_width` in your thesis' `index.Rmd` file:

```
params:
  other_path: "../paper_to_include/"
  out_width: "80%"
```

Now, the output width of your figure will be 80 % when knitting your paper on its own, and 100 % when knitting it as child document of your thesis.

5.11. Customizing citations and referencing

5.11.1. Using a .csl file with pandoc

See section [3.1.1](#).

The only drawbacks to letting pandoc handle citations is that (i) it does not support chapter bibliographies, (ii) if you're a LaTeX veteran, you might be more comfortable with `biblatex` or `natbib`.

5.11.2. Using biblatex

To use `biblatex` to handle citations, first uncomment this in `index.Rmd`, YAML header:

```
use-biblatex: true
bib-latex-options: "style=authoryear, sorting=nyt, backend=biber,
↪ maxcitenames=2, useprefix, doi=true, isbn=false,
↪ uniquename=false"
```

Then tell R Markdown to use `biblatex` when inserting citations, by setting `citation_package: biblatex`:

```
output:
  bookdown: pdf_book:
    citation_package: biblatex
```

To customise the appearance of citations, change `bib-latex-options`. For example, to get **numerical citations**, with references in order of their appearance in the text, set it to

```
bib-latex-options: "style=numeric-comp, sorting=none,
↪ backend=biber, maxcitenames=2, useprefix, doi=true, isbn=false,
↪ uniquename=false"
```

Adding chapter bibliographies

If you would like chapter bibliographies, first add “`refsection=chapter`” to the `biblatex` options, for example like this:

5. Customisations and extensions

```
bib-latex-options: "refsection=chapter, style=authoryear,  
↪  sorting=nyt, backend=biber, maxcitenames=2, useprefix,  
↪  doi=true, isbn=false, uniquename=false"
```

Second, set the parameter `insertHeadingInPDF: false` in `index.Rmd`, to suppress the inclusion of a ‘References’ heading at the end of the thesis.

```
params:  
  insertHeadingInPDF: false
```

Finally insert this line at the end of each chapter, to print the bibliographies there:

```
\printbibliography[segment=\therefsection,heading=subbibliography]
```

5.11.3. Using `natbib`

To use `natbib` to handle citations, first uncomment this in `index.Rmd`, YAML header:

```
use-natbib: true  
natbib-citation-style: authoryear #for science, you might want  
↪  numbers,square  
natbib-bibliography-style: templates/ACM-Reference-Format.bst #e.g.  
↪  "plainnat", unsrtnat, or path to a .bst file
```

Then tell R Markdown to use `natbib` when inserting citations, by setting `citation_package: natbib`:

```
output:  
  bookdown: pdf_book:  
    citation_package: natbib
```

To customise the appearance of citations, change what `.bst` file you point to in `natbib-bibliography-style`.

5.12. Customizing the page headers and footers (PDF)

This can now be done directly in **index.Rmd**'s YAML header. If you are a LaTeX expert and need further customisation that what's currently provided, you can tweak the relevant sections of **templates/template.tex** - the relevant code is beneath the line that begins `\usepackage{fancyhdr}`.

5.13. Diving in to the OxThesis LaTeX template (PDF)

For LaTeX minded people, you can read through **templates/template.tex** to see which additional customisation options are available as well as **templates/ociamthesis.cls** which supplies the base class. For example, **template.tex** provides an option for master's degree submissions, which changes identifying information to candidate number and includes a word count. At the time of writing, you must set this directly in **template.tex** rather than from the YAML header in **index.Rmd**.

5.14. Customising institutional details

5.14.1. The minimal route

Customising to your institutional needs is as simple as putting the name of your institution and the path to your university's logo in **index.Rmd**:

```
university: Universidad de las Fuerzas Armadas ESPE
university-logo: templates/beltcrest.pdf
```

5.14.2. Replacing the entire title page with your required content

If you have a **.tex** file with some required front matter that you want to replace the default title page altogether, you can provide a filepath to this file

5. Customisations and extensions

in `index.Rmd`. The sample content includes an example of this — if you use the YAML below, your front matter will look like this:

alternative-title-page:
↪ front-and-back-matter/alt-title-page-example.tex

| | | |
|--|---|--|
| <div>Title of your Thesis</div> <div>John Doe</div> | <div><div>Thesis committee</div><div><div>Promotor:</div><div>Prof.dr. J. Smith
Professor of Geo-information Science and Remote Sensing
Wageningen University</div><div><div>Co-promotors:</div><div>Dr. Name of co-promotor
Assistant Professor, Laboratory of Geo-information Science and Remote Sensing
Wageningen University</div><div><div>Other members:</div><div>Prof.dr. Jury member 1, Wageningen University
Prof.dr. Jury member 2, Affiliation
Prof.dr. Jury member 3, Affiliation
Prof.dr. Jury member 4, Affiliation</div></div><div>This research was conducted under the auspices of the C.T. de Wit Graduate School of Production Ecology & Resource Conservation (PERC)</div></div></div></div> | <div><div>Title of your thesis</div><div>John Doe</div></div> <div><div>Thesis</div><div>submitted in fulfillment of the requirements for the degree of doctor at
Wageningen University
by the authority of the Rector Magnificus
Prof. Dr A.P.J. Mol,
in the presence of this
Thesis Committee appointed by the Academic Board
to be defended in public
on Date of your defense
at 4 p.m. in the Aula</div></div> |
| <div>John Doe
Title of your thesis
77 pages
PhD thesis, Wageningen University, Wageningen, NL (2015)
With references, with summary in English
ISBN XXX-YYY</div> | <div>For Yihui Xie</div> | <div>Acknowledgements</div> <div><p>This is where you will normally thank your advisor, colleagues, family and friends, as well as funding and institutional support. In our case, we will give our praises to the people who developed the ideas and tools that allow us to push open science a little step forward by writing plain-text, transparent, and reproducible theses in R Markdown.</p><p>We must be grateful to John Graham for inventing the original version of Markdown, to John MacFarlane for creating Pandoc (http://pandoc.org) which converts Markdown to a large number of output formats, and to Yihui Xie for creating knitr which introduced R Markdown as a way of embedding code in Markdown documents, and bookdown which added tools for technical and longer-form writing.</p><p>Special thanks to Christine Loney, who created the thesistemplate package that helped many a PhD student write their theses in R Markdown. And a very special thanks to John McManis, whose adoption of Sam Evans' adaptation of Keith Gillen's original auctex template for writing an Oxford University DPhil thesis in LaTeX provided the template that I in turn adopted for R Markdown.</p><p>Finally, profuse thanks to JJ Allaire, the founder and CEO of RStudio, and Hedy Wickham, the mastermind of the tidyverse without whom we'd all just given up and done data science in Python instead. Thanks for making data science easier, more accessible, and more fun for us all.</p><div>Ulrik Lenga
Linacre College, Oxford
2 December 2018</div></div> |

6

Troubleshooting

This chapter describes common errors you may run into, and how to fix them.

6.1. Error: Failed to build the bibliography via biber

This can happen if you've had a failed build, perhaps in relation to RStudio shutting down abruptly.

Try doing this:

1. type `make clean-knits` in the terminal tab (or run `file.remove(list.files(pattern = ".*(log|mtc|maf|aux|bbl|blg|xml)"))` in the R console) to clean up files generated by LaTeX during a build
2. restart your computer

If this does not solve the problem, try using the [natbib](#) LaTeX package instead of [biblatex](#) for handling references. To do this, go to **index.Rmd** and

1. set `use-biblatex: false` and `use-natbib: true`
2. set `citation_package: natbib` under

6. *Troubleshooting*

```
output:  
  bookdown::pdf_book:  
    citation_package: natbib
```

*Alles Gescheite ist schon gedacht worden.
Man muss nur versuchen, es noch einmal zu denken.*
*All intelligent thoughts have already been thought;
what is necessary is only to try to think them again.*
— Johann Wolfgang von Goethe (Goethe, 1829)

Conclusion

If we don't want Conclusion to have a chapter number next to it, we can add the `{-}` attribute.

More info

And here's some other random info: the first paragraph after a chapter title or section head *shouldn't be* indented, because indents are to tell the reader that you're starting a new paragraph. Since that's obvious after a chapter or section title, proper typesetting doesn't add an indent there.

This paragraph, by contrast, *will* be indented as it should because it is not the first one after the 'More info' heading. All hail LaTeX. (If you're reading the HTML version, you won't see any indentation - have a look at the PDF version to understand what in the earth this section is babbling on about).

Appendices



Primer Apéndice

Este primer apéndice incluye bloques de código R complementarios que fueron configurados para enriquecer la lectura del documento:

Estructura del bloque de código

```
library(tidyverse)
knitr::include_graphics("figures/sample-content/chunk-parts.png")
```

Ejemplo de transformación de gráfico/imagen del Capítulo 2.2:

```
knitr::include_graphics("figures/sample-content/beltcrest.png")
```

B

Segundo Apéndice: Notas Metodológicas y Computacionales

En este apéndice se registran las especificaciones del entorno computacional utilizado para garantizar la total reproducibilidad de los resultados y estimaciones econométricas.

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